Proof of (21.3.67)

Proof that $\mathrm{Tr}\, A = \int \langle -\bar{\psi} | A | \psi \rangle e^{-\bar{\psi}\psi} d\bar{\psi} d\psi$

$$\mathrm{Tr}\, A = \sum_n \langle n | A \underset{\wedge}{} | n \rangle =$$

$$\hat{1} = \int |\psi\rangle\langle\bar{\psi}| e^{-\bar{\psi}\psi} d\bar{\psi} d\psi$$

$$= \sum_n \int \langle n | A | \psi \rangle \langle \bar{\psi} | n \rangle e^{-\bar{\psi}\psi} d\bar{\psi} d\psi =$$

$$= \sum_n \int \langle n | A \left( |0\rangle - \psi |1\rangle \right) \left( \langle 0| - \langle 1| \bar{\psi} \right) | n \rangle e^{-\bar{\psi}\psi} d\bar{\psi} d\psi$$

$$= \sum_n \int \left( \langle n | A | 0 \rangle + \langle n | A | 1 \rangle \psi \right) \left( \langle 0 | n \rangle + \bar{\psi} \langle 1 | n \rangle \right) \cdot$$

$$\cdot e^{-\bar{\psi}\psi} d\bar{\psi} d\psi$$

$$= \sum_n \int \Big\{ \langle n | A | 0 \rangle \langle 0 | n \rangle + \langle n | A | 1 \rangle \psi \langle 0 | n \rangle$$

$$+ \langle n | A | 0 \rangle \bar{\psi} \langle 1 | n \rangle + \langle n | A | 1 \rangle \psi \bar{\psi} \langle 1 | n \rangle \Big\} \cdot$$

$$\cdot (1 - \bar{\psi}\psi) d\bar{\psi} d\psi = \text{(drop all integrals that are zero)} =$$

$$= \sum_n \int \langle n | A | 0 \rangle \langle 0 | n \rangle \left( -\bar{\psi}\psi d\bar{\psi} d\psi \right)$$

$$+ \langle n | A | 1 \rangle \psi \bar{\psi} \langle 1 | n \rangle \left( 1 - \bar{\psi}\psi \right) d\bar{\psi} d\psi$$

Use now $\begin{cases} \iint -\bar{\psi}\psi d\bar{\psi} d\psi = \int \bar{\psi} d\bar{\psi} \, \psi d\psi = 1 \\ \int \psi \bar{\psi} d\bar{\psi} d\psi = 1 \end{cases}$

$$\mathrm{Tr}\, A = \sum_n \langle n | A | 0 \rangle \langle 0 | n \rangle + \langle n | A | 1 \rangle \langle 1 | n \rangle = \langle 0 | A | 0 \rangle + \langle 1 | A | 1 \rangle.$$

On the other hand, consider

$$\int \langle -\bar{\psi} | A | \psi \rangle e^{-\bar{\psi}\psi} d\bar{\psi} d\psi = \left( \langle \bar{\psi} | = \langle 0 | - \langle 1 | \bar{\psi} = \langle 0 | + \bar{\psi} \langle 1 | \right)$$

$$= \int \left( \langle 0 | - \bar{\psi} \langle 1 | \right) A \left( \underset{|0\rangle + |1\rangle \psi}{|0\rangle - \psi |1\rangle} \right) (1 - \bar{\psi}\psi) \, d\bar{\psi} d\psi =$$

$$= \int \Big( \langle 0 | A | 0 \rangle + \langle 0 | A | 1 \rangle \psi - \bar{\psi} \langle 1 | A | 0 \rangle - \bar{\psi} \langle 1 | A | 1 \rangle \psi \Big) (1 - \bar{\psi}\psi) \, d\bar{\psi} d\psi =$$

(All matrix elements $\langle l | A | k \rangle$ with $l = 0, 1$; $k = 0, 1$ are just c-numbers)

$$= \langle 0 | A | 0 \rangle \underbrace{\int (-\bar{\psi}\psi) \, d\bar{\psi} d\psi}_{1} + \langle 0 | A | 1 \rangle \underbrace{\int \psi (1 - \bar{\psi}\psi) \, d\bar{\psi} d\psi}_{0}$$

$$- \langle 1 | A | 0 \rangle \underbrace{\int \bar{\psi} (1 - \bar{\psi}\psi) \, d\bar{\psi} d\psi}_{0}$$

$$- \langle 1 | A | 1 \rangle \int \bar{\psi}\psi (1 - \bar{\psi}\psi) \, d\bar{\psi} d\psi =$$

$$= \langle 0 | A | 0 \rangle + \langle 1 | A | 1 \rangle \underbrace{\int (-\bar{\psi}\psi)(1 - \bar{\psi}\psi) \, d\bar{\psi} d\psi}$$

$$\int (-\bar{\psi}\psi) \, d\bar{\psi} d\psi$$

$$+ \int (\bar{\psi}\psi)(\bar{\psi}\psi) \, d\bar{\psi} d\psi =$$

$$= 1 - \int \psi \underbrace{\bar{\psi}\bar{\psi}}_{\bar{\psi}^2 = 0} \psi \, d\bar{\psi} d\psi = 1$$

$$= \langle 0 | A | 0 \rangle + \langle 1 | A | 1 \rangle .$$

Thus $\mathrm{Tr}\, A = \int \langle -\Psi | A | \Psi \rangle e^{-\bar{\Psi}\Psi} d\bar{\Psi} d\Psi$

for any bosonic operator A (that is, any operator made of even number of Fermi operators)

Note that integral like $\int \langle \bar{\Psi} | A | \Psi \rangle e^{-\bar{\Psi}\Psi} d\bar{\Psi} d\Psi$

would give $\langle 0 | A | 0 \rangle - \langle 1 | A | 1 \rangle$ instead!